3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius *r*, as shown in Fig. 3.1.

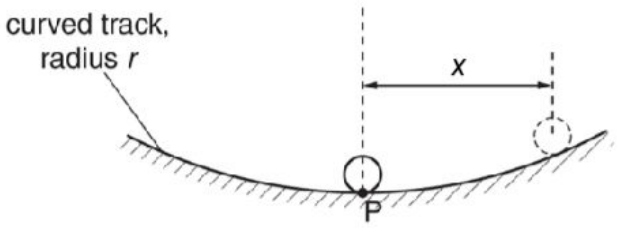


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

Since *g*/*r* is constant, *a* is proportional to *x*. [1]

The minus sign shows that *a* is opposite to *x*. [1]

.....[2]

(ii) The radius *r* of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

Compare given equation with $a = -\omega^2 x$,

$\omega^2 = \frac{g}{r}$ [1]

$(\frac{2\pi}{T})^2 = \frac{9.81}{28\times10^{-2}}$

T = 1.06 s [1]

Time taken between ball passing point P and its first return to point P

= T/2 = 1.06/2 = 0.53 s [1]

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

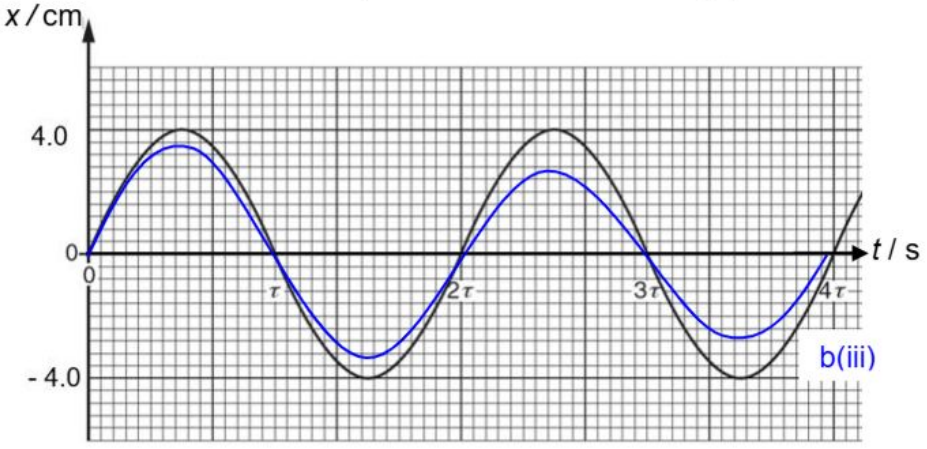


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

$$KE_{max} = \frac{1}{2} m\omega^2 x_o^{\ 2}$$

$$KE_{max} = \frac{1}{2}(10\times10^{-3})(\frac{9.81}{28\times10^{-2}})(4.0\times10^{-2})^2 \text{ [1]}$$

$$KE_{max} = 2.8\times10^{-4} \text{ J [1]}$$

maximum kinetic energy= ..... J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

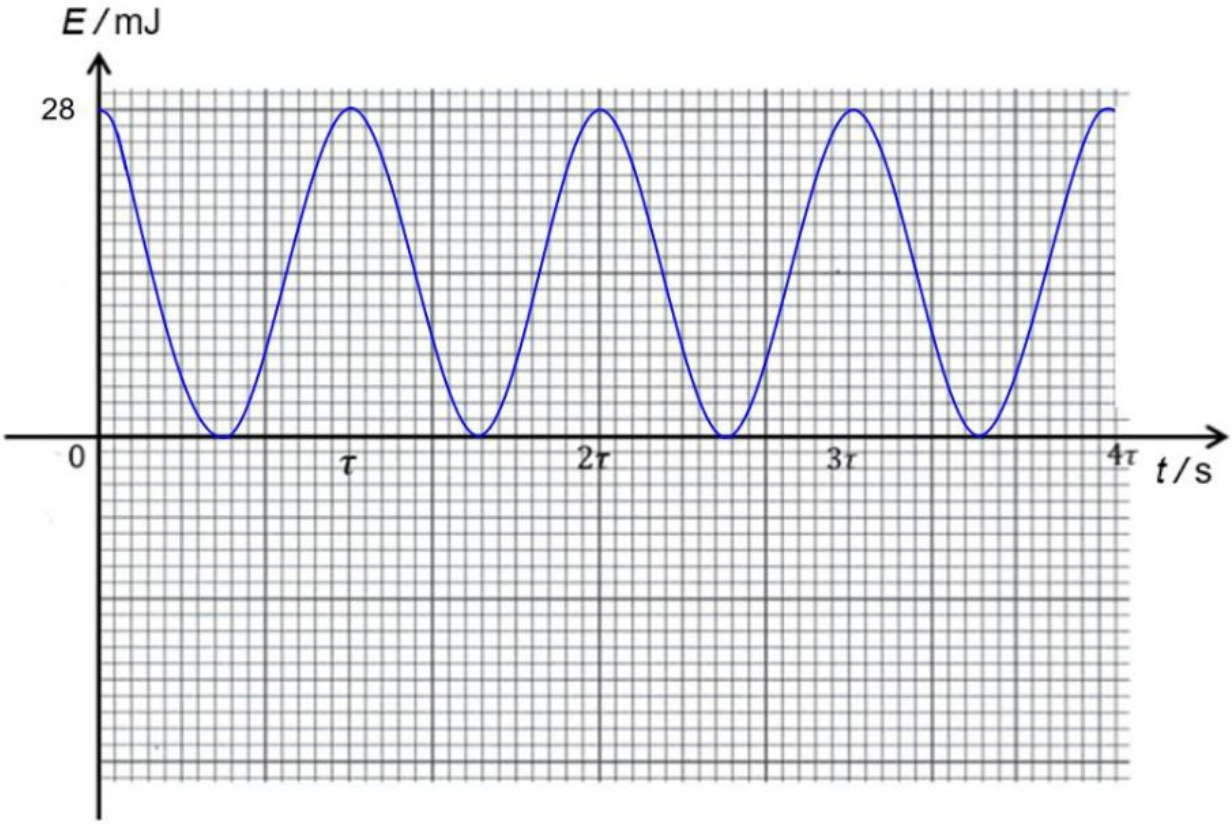


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

Amplitude drops with time, period hardly changes.

[Total: 9]